

Assignment 4

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Problem (Munkres 23.10). Let $\{X_\alpha\}_{\alpha \in J}$ be an indexed family of connected spaces; let X be the product space

$$X = \prod_{\alpha \in J} X_\alpha.$$

Let $a = (a_\alpha)$ be a fixed point of X .

- (a) Given any finite subset K of J , let X_K denote the subspace of X consisting of all points $x = (x_\alpha)$ such that $x_\alpha = a_\alpha$ for $\alpha \notin K$. Show that X_K is connected.
- (b) Show that the union Y of the spaces X_K is connected.
- (c) Show that X equals the closure of Y . Conclude that X is connected.

Proof of (a). $K \subseteq J$ is a finite subset, say

$$K = \{\alpha_1, \dots, \alpha_n\}.$$

Then $x = (x_\alpha) \in X_K$ coincides with a on all, but finitely many x_α , namely the indices

$$\{x_{\alpha_i}\}_{i \in [n]}.$$

More importantly, $x, y \in X_K$ coincide on all but these finitely many indices.

The most straightforward approach for connectedness is to show that

$$X_K \cong \prod_{i \in [n]} X_{\alpha_i}.$$

Connectedness is preserved under homeomorphism.

Let $\varphi : X_K \rightarrow \prod_{i \in [n]} X_{\alpha_i}$ be the map that sends

$$(x_\alpha) \rightarrow (x_{\alpha_1}, x_{\alpha_2}, \dots, x_{\alpha_n}).$$

Since, for all $x, y \in X_K$, they coincide on all but these finitely many indices.

Thus there is a natural inverse $\varphi^{-1} : \prod_{i \in [n]} X_K \rightarrow X_K$ which sends $(x_{\alpha_1}, \dots, x_{\alpha_n})$ to the $x = (x_\alpha) \in X_K$ for which the finitely many nonfixed indices are $x_{\alpha_1}, x_{\alpha_2}, \dots, x_{\alpha_n}$.

For rigor, define the injection $\iota : [n] \rightarrow J$ which sends i to the index in J corresponding to α_i .

Then φ^{-1} is the function mapping

$$(x_{\alpha_1}, \dots, x_{\alpha_n}) \mapsto y = (y_\alpha)_{\alpha \in J}$$

where $y_{\iota_i} = x_{\alpha_i}$ and $y_\alpha = a_\alpha$ otherwise.

This map is well-defined since $y = \varphi^{-1}(x_{\alpha_1}, \dots, x_{\alpha_n})$ coincides with a on all, but the finitely many indices.

It is not terribly hard to see that $\varphi^{-1} \circ \varphi \equiv \text{id}_{K_X}$ and that $\varphi \circ \varphi^{-1} \equiv \text{id}_{\prod X_\alpha}$.

It remains to prove that these maps are continuous.

Let $U \subseteq \prod_{i \in [n]} X_{\alpha_i}$ be an open set.

Then,

$$U = \bigcup_{\beta} \prod_{i=1}^n U_i, \quad U_i \text{ open in } X_{\alpha_i},$$

which is annoying.

Luckily, it suffices to show the preimage of basis elements are open in the domain.

The basis elements of the topology on $\prod_{i \in [n]} X_{\alpha_i}$ are

$$\prod_{i=1}^n U_i, \quad U_i \text{ open in } X_{\alpha_i},$$

and the ones in X_K are

$$\prod_{\alpha} U_{\alpha},$$

where U_{α} are open in X_K and only finitely many $U_{\alpha} \neq X_K$.

For a basis element $\prod U_i$ of $\prod_i X_{\alpha_i}$, we have

$$\varphi^{-1} \left(\prod_{i=1}^n U_i \right) = \prod_{\alpha} U_{\alpha}$$

where only those U_{β} for which

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